

Consider the polyhedron defined as follows:

$$\mathcal{P} = \{x \in \mathbb{R}^2 \mid -x_1 + x_2 \leq 2, x_1 + x_2 \leq 4, x_1 \geq 0, x_2 \geq 0\}.$$

Determine whether the given directions $\tilde{d} \in \mathbb{R}^2$ are feasible for the given feasible points $\tilde{x}$ of $\mathcal{P}$:

1. $\tilde{x}_a = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \tilde{d}_a = \begin{pmatrix} 1 \\ 0 \end{pmatrix};$
2. $\tilde{x}_b = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \tilde{d}_b = \begin{pmatrix} -1 \\ 0 \end{pmatrix};$
3. $\tilde{x}_c = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \tilde{d}_c = \begin{pmatrix} 0 \\ 2 \end{pmatrix};$
4. $\tilde{x}_d = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \tilde{d}_d = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$